

AGGREGATE TRENDS

PRESENTATION 5

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Wednesday, December 10



Front Desk Analysis



Callers Reach the Front Desk Through Four Primary Routes

Transfer Type	Definition
Front Desk Transfer (FDT)	Accessed through the Legal Menu pathway (LegalMenu1 or LegalMenu2) for callers who need interpreter services.
Front Desk Transfer 1 (FDT 1)	Triggered directly from the Language Selection Menu when callers do not press 1 (English) or 2 (Spanish). These callers either need interpreter services, are waiting as instructed by the prompt, or are unsure how to proceed.
Front Desk Transfer 2 (FDT 2)	Accessed after callers navigate through specific menu options, most commonly Help With Legal or Other Reason Menu (55%), Appointment Menu (41%), or Compliment Or Complaint Menu (4%).
FrontDeskTransfer3 (FDT 3)	An automated timeout mechanism triggered after callers experience on average 3 Main Menu repetitions at exactly 60-second intervals without making a selection.

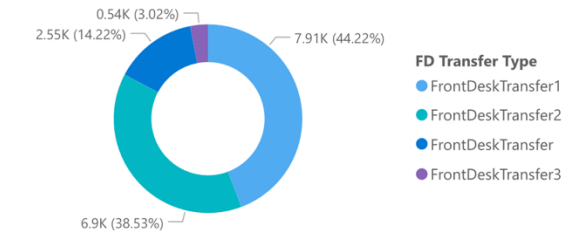
Dashboard Walk Through

The link can be found [here](#)

Front Desk Analysis

FrontDeskTransfer (FDT): Accessed through the Legal Menu pathway (LegalMenu1 or LegalMenu2) for callers who need interpreter services.
FrontDeskTransfer1 (FDT1): Triggered directly from the Language Selection menu when callers do not press 1 (English) or 2 (Spanish) because they need interpreter services.
FrontDeskTransfer2 (FDT2): Accessed after callers navigate through specific menu options: Help With Legal or Other Reason Menu, Appointment Menu or Compliment/Complaint Menu.
FrontDeskTransfer3 (FDT3): An automated timeout mechanism that is triggered after callers experience on average 3 Main Menu repetitions without making a selection.

Call Volume by FD Transfer Type



Average Call Duration by Transfer Type

Time spent talking to FD staff (excludes menu navigation time)

Front Desk Type	Average Duration (min)	Median Duration (Min)
FrontDeskTransfer	5.01	4.37
FrontDeskTransfer1	2.99	1.27
FrontDeskTransfer2	4.10	2.92
FrontDeskTransfer3	2.05	0.27

1,096.1 hours

Total Staff Time on Front Desk Calls

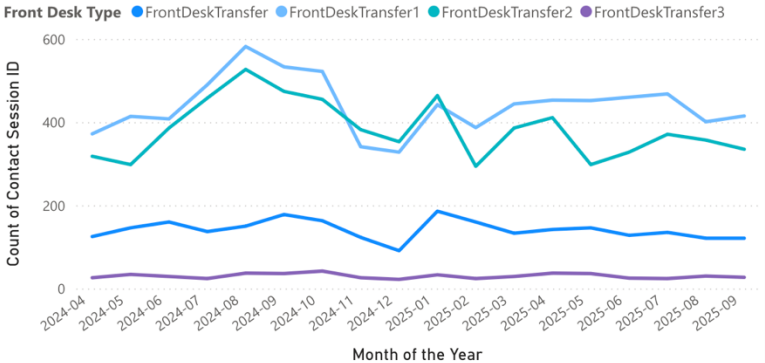
3.1 hours

Average Daily Staff Time on Front Desk Calls

Time Parameter

Month of the Year

Call Volume by Time Parameter



Front Desk Staff Have Significant Available Capacity

Hourly volumes divided by 370 weekdays for avg. daily operational demand

Hour	FDT	FDT 1	FDT 2	FDT 3	Total Daily
8:00 AM	0.24	0.53	0.52	0.02	1.31
9:00 AM	0.96	2.93	2.72	0.20	6.81
10:00 AM	1.05	2.97	2.57	0.21	6.80
11:00 AM	1.06	3.02	2.62	0.21	6.91
12:00 PM	0.84	2.58	2.36	0.20	5.98
1:00 PM	0.89	2.63	2.25	0.18	5.95
2:00 PM	0.83	2.60	2.19	0.16	5.78
3:00 PM	0.73	2.54	1.96	0.16	5.39
4:00 PM	0.29	1.59	1.45	0.12	3.45

1,096.1 hours

Total Staff Time on Front Desk Calls

3.1 hours

Average Daily Staff Time on Front Desk Calls

- **Peak Hours are 9-11:00 AM** with ~ 7 front desk calls per day average
- Front desk **handles 1 call every 8-9 minutes** during peak hours
- Average call duration 3-5 minutes = **~ 25-35 minutes of talk time per peak hour**
- 3 hours (on average) daily commitment for front desk calls means **62% of an 8-hour day is available for other client services**

Most Front Desk Calls Resolve Quickly Despite Some Higher Averages

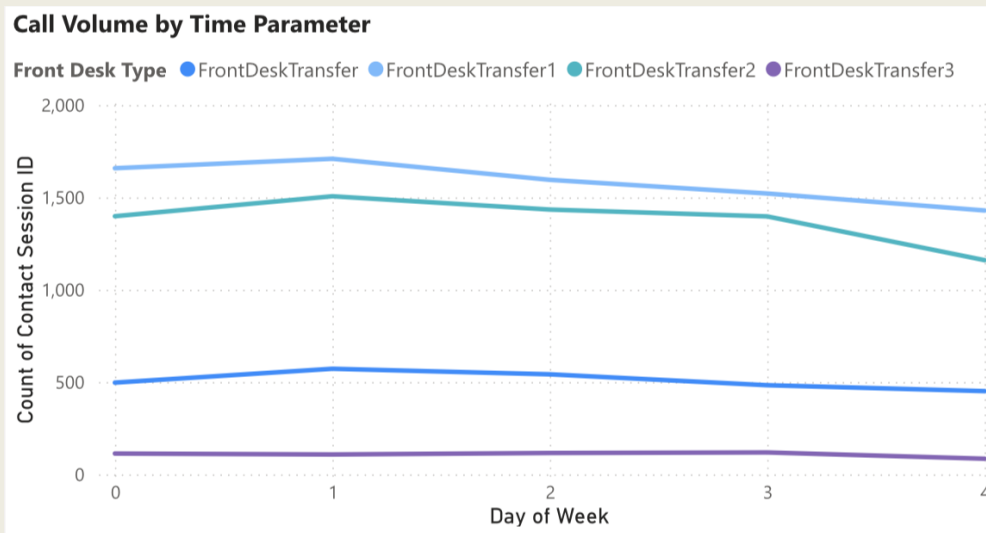
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- **FDT1** (language selections) calls have the **shortest duration** outside of FDT3
- **FDT3** (timeout after menu repetitions) **resolve very quickly**
- **FDT2** (menu navigation) **shows the biggest variation:** simple directions are quick, complex menu explanations take longer
- **Averages are inflated by outliers:** most front desk interactions are resolved in **under 3 minutes**

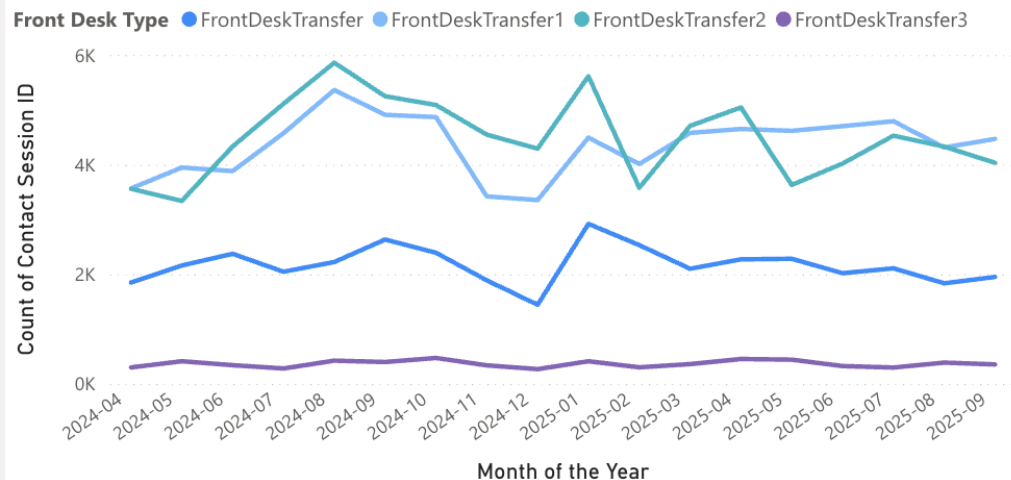
Tuesday Shows Consistent Volume Spike Across the Main Three Call Paths



- Tuesday **peaks across all main call pathways** with FDT increasing by 15% and FDT2 increasing by 8%
- **Friday shows consistent decline** indicating end-of-week demand patterns
- There are **predictable staffing needs** that can be **planned for operationally**

The Predictable Nature of Seasonal Fluctuations Could Help Adjust Staffing

Call Volume by Time Parameter



- **Summer peak (July – August 2024):** highest volume across most categories (5,361 FDT1 calls in August vs. ~ 4000 on average)
- **Holiday dips (November – December 2024):** show 20-25% volume decline
- **January volume increase is expected** and post-holiday rebound allows for front desk **staffing to be adjusted proactively**

Recommendations

1 Pathway Specific



Front Desk Transfer 1 (Language Menu)

- Add third popular language: Polish, Arabic, or Russian
- Add “Press 0 for interpreter services”

Front Desk Transfer (Legal Menus)

- Opportunity to consolidate legal menu pathways
(detailed analysis by Liv & Ella)

Front Desk Transfer 2 & 3

- No recommendations needed

2 Staffing & Capacity



Expand Front Desk Access in Menu

- Add “Press 0 for assistance” or other prompts in select places of menu system with known bottlenecks

Leverage Available Staff Capacity

- There is up to 61% of staff time in a working day not spent talking to clients
- Opportunity to expand responsibilities beyond just call routing

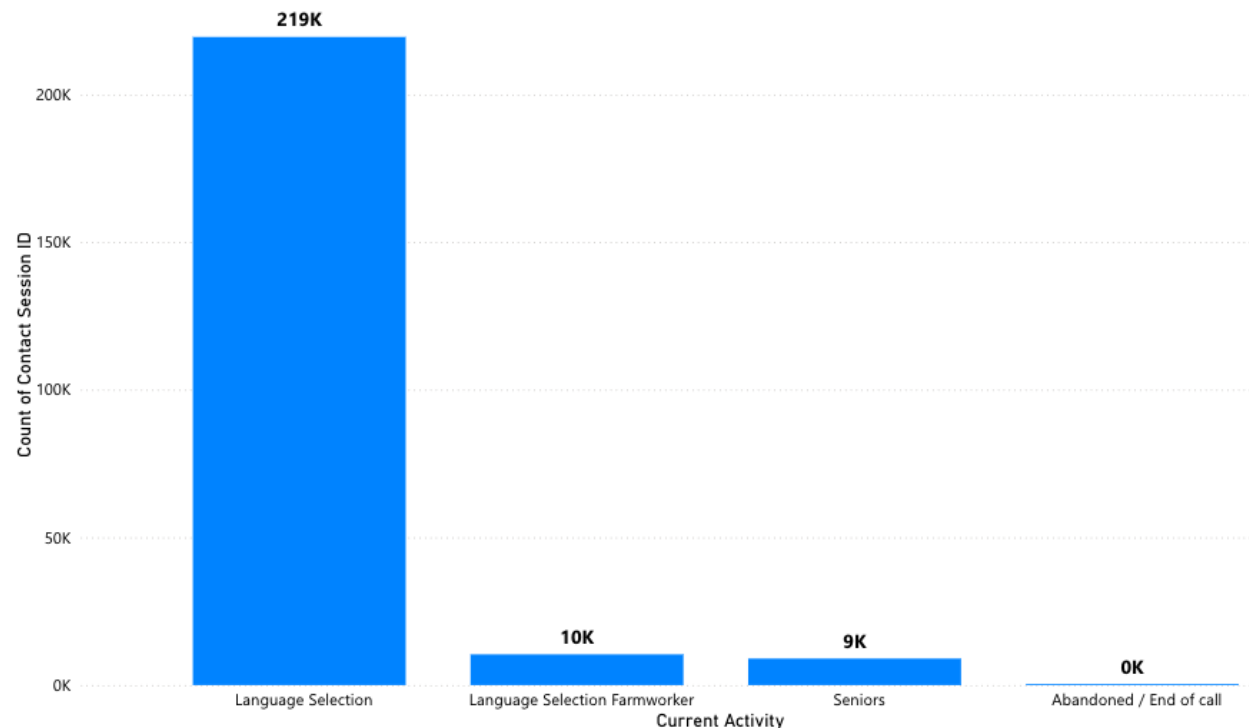
Submenu Analysis

Hierarchical Menu Usage Analysis: Power BI Drill-Down Dashboard



Finalized Dashboard

Count of Contact Session ID by Current Activity

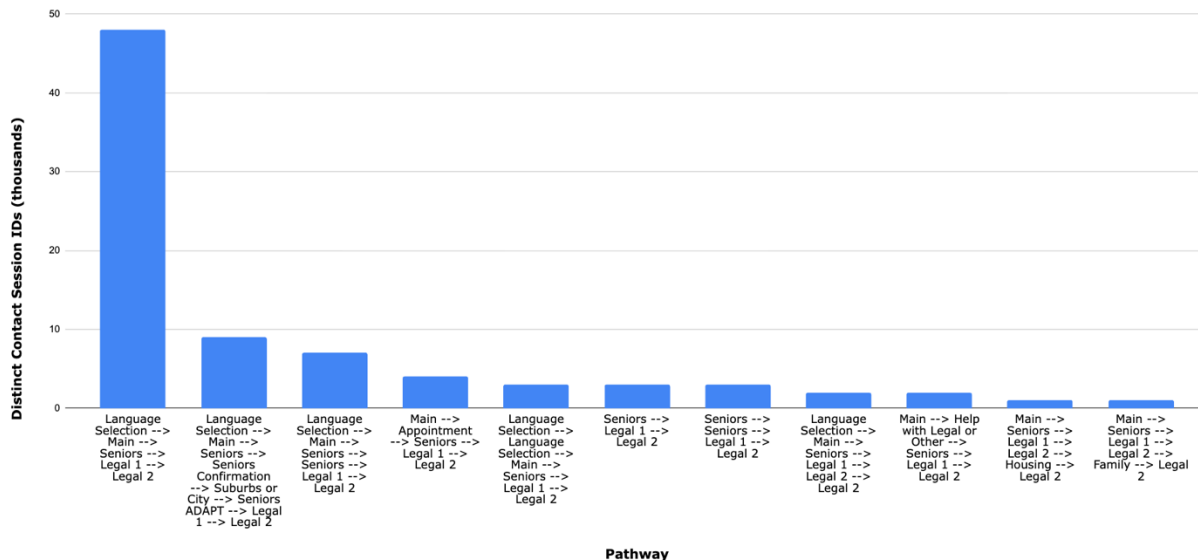


What We Did:

- Added the Farmworker Dashboard to the Main one
- Expanded the Dashboard to reflect up to 22 activities
- Updated the topmost tier to have 4 potential starting points

Legal Menu 2 Access is Widely Spread

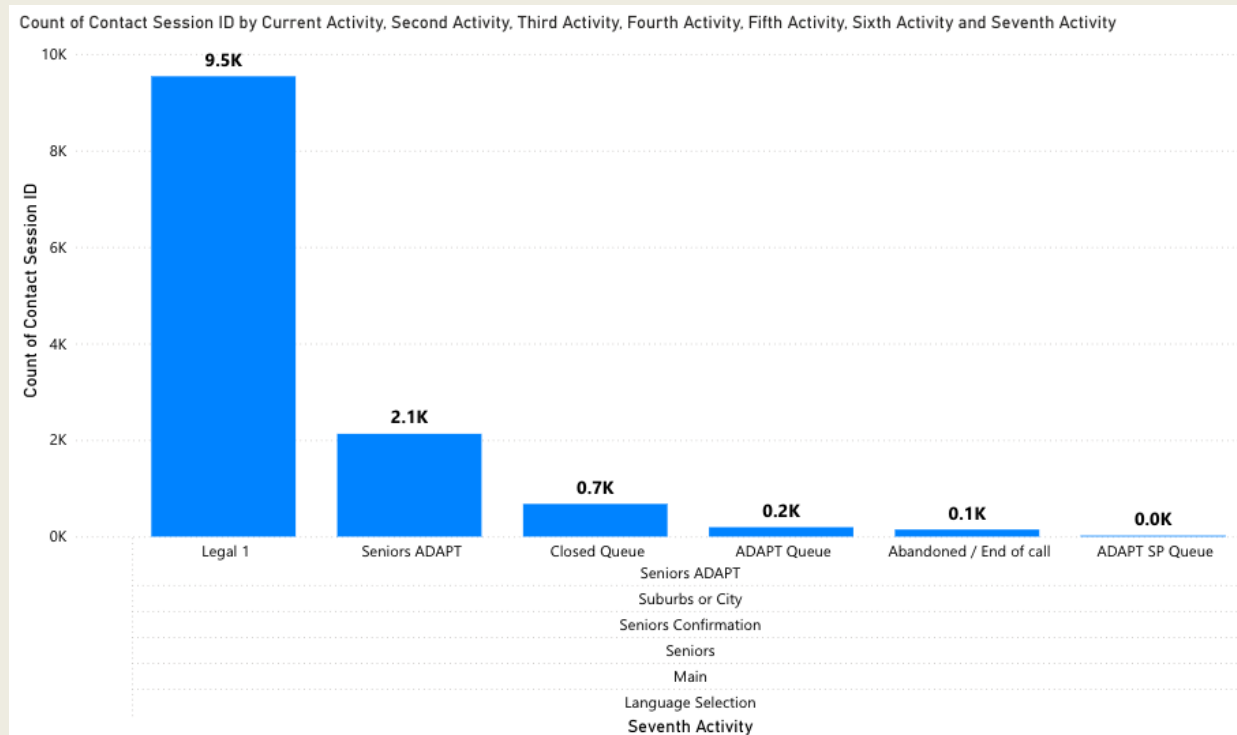
Spread of Legal Menu 2 Access



Insights:

- LegalMenu1 functions almost entirely as a pass-through
- ~37k callers reach LegalMenu2 through repetitive or looping routes
- These patterns suggest callers are taking unnecessarily long paths that don't affect the final routing decision

Chicago Seniors Don't Have ADAPT Issues



Insights:

- ~75% of people who go to Seniors ADAPT go straight to Legal Menu 1
- ~15% go back to Seniors ADAPT and then go to Legal Menu 1
- Only ~1.6% of callers who go to Seniors ADAPT enter the ADAPT Queue

Recommendations

1

Remove Legal Menu 1

- Almost all callers who go to Legal Menu 1 then go to Legal Menu 2
- Combine Legal Menu 1 and Legal Menu 2 into a single Legal Menu

2

Legal Menu 2 should be consolidated

- Only 48k out of 85k callers reach Legal Menu 2 the most direct way

3

Remove Seniors ADAPT as an option from Suburbs or City

- Because ~75% of callers in Seniors ADAPT go straight to Legal Menu 1, Legal Menu 1 should replace Seniors ADAPT as a pathway from the Suburbs or City Menu

Appendix

** Includes additional details and visuals from our analysis that were not included in the live presentation due to time constraints.*

Appendix A

The following data analysis was completed in **Python** for **CAR Dataset**. Tatum's full code script is [Here](#)

```
In [8]: import pandas as pd
import numpy as np

# Read the main data
df_main = pd.read_csv("combined_calls_CAR.csv")
df_main['Activity Start Timestamp'] = pd.to_datetime(df_main['Activity Start Timestamp'])
df_main = df_main.sort_values(['Contact Session ID', 'Activity Start Timestamp'])

print("Creating comprehensive front desk dataset...")

# =====
# 1. IDENTIFY FRONT DESK SESSIONS AND ADD FLAGS
# =====

# Mark front desk activities
df_main['is_front_desk'] = df_main['Activity Name'].str.contains('FrontDesk', case=False, na=False)

# Get sessions with front desk interactions
fd_sessions_list = df_main[df_main['is_front_desk']]['Contact Session ID'].unique()

# Add session-level flags
df_main['has_front_desk'] = df_main['Contact Session ID'].isin(fd_sessions_list)

# =====
# 2. GET FRONT DESK TYPE FOR EACH SESSION
# =====

# Get the front desk type for each session (first occurrence)
fd_types = df_main[df_main['is_front_desk']].groupby('Contact Session ID')['Activity Name'].first()

# Merge back to main dataframe
df_main = df_main.merge(
    fd_types.rename('session_fd_type').reset_index(),
    on='Contact Session ID',
    how='left'
)

# =====
# 3. ADD TIME COMPONENTS FOR FRONT DESK SESSIONS
# =====

# Get first front desk timestamp for each session
```